

HACK THE WORLD · WORLD-MODELS TRACK

Two-Rooms World Model

An action-conditioned JEPA for goal-reaching navigation

Team **Organizer** · June 2026 · example submission

Color code in this deck — red = a possible improvement gold = an exploration possibility

BEFORE WE START

How to read this deck

▢ **It's a structure.** A clear, section-by-section way to **present the results** of your study.

💡 **It's a source of ideas.** The red & gold notes suggest ways to **enrich your analysis and presentation.**

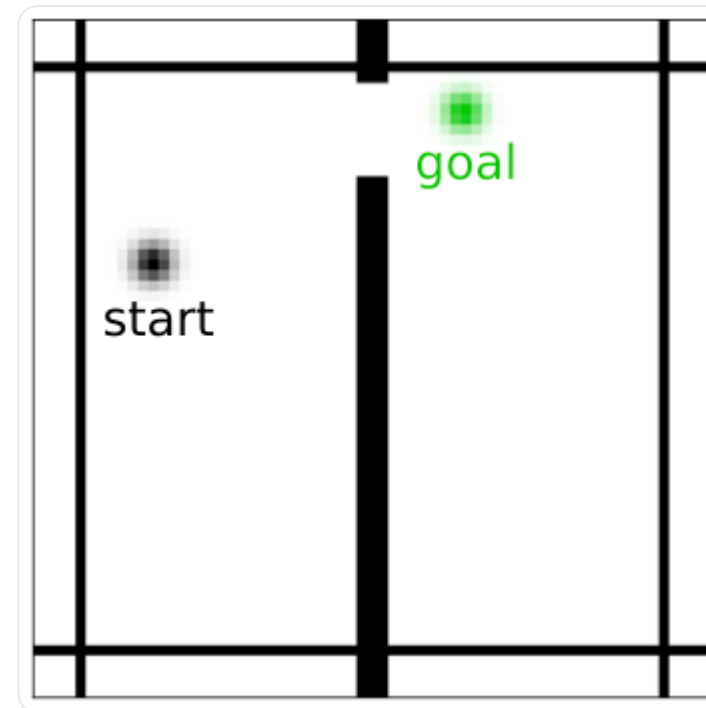
✗ WHAT NOT TO DO

Don't try to implement everything suggested here. The core of your work should be **your own research** — searching for new regularizations, borrowing methods from other fields, trying fresh ideas. These notes are only about **presenting good results excellently** — moving a good submission to an outstanding one.

Goal-reaching in Two Rooms

- A dot navigates **two rooms joined by a door** — the DotWall env.
- **Inputs:** pixel observations + actions.
- **Learn** a world model whose latent predicts future states.
- **Then plan** actions to reach a goal position.

◇ **COULD HELP** Framing the whole submission as a **scaling study** rather than a single run would be far more convincing than one number.



A single two_rooms episode: start → goal across the door.

The example pipeline

Example submission = baseline EB-JEPA, unchanged,
trained on two_rooms via the ac_video_jepa recipe.

1 · Data generation

two_rooms rollouts



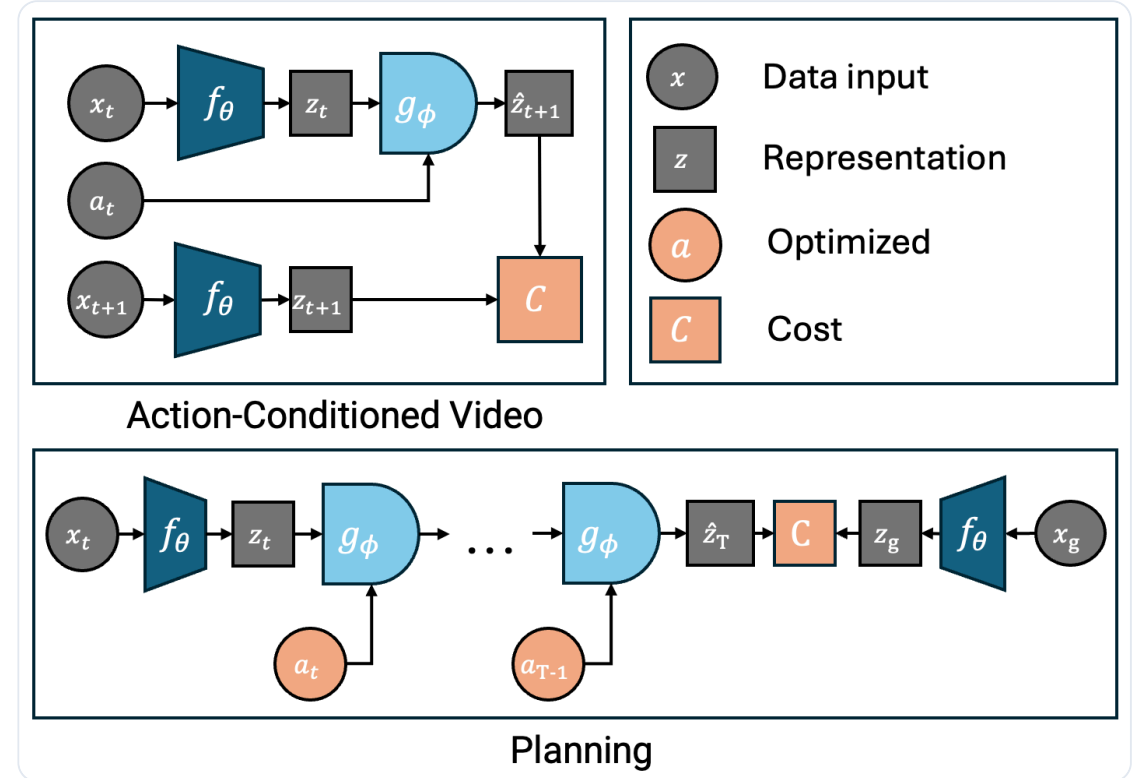
2 · JEPA training

encoder + predictor + action module



3 · Evaluation

actor plans to the goal



Base EB-JEPA: action-conditioned training (top) & planning unroll (bottom).

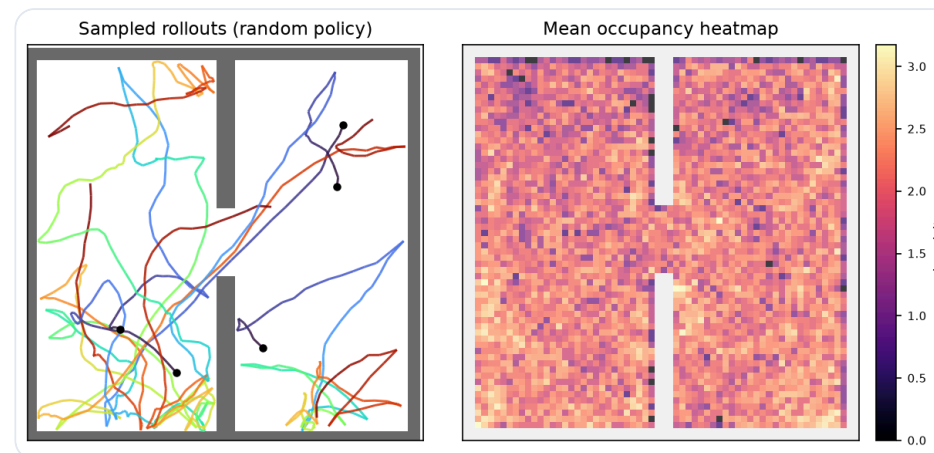
Dataset selection

Generated (synthetic). two_rooms trajectories produced on-the-fly.

- **Methodology:** procedural wall/door layout + random-action rollouts
- **Modality:** `online` GPU `stream` pipeline (double-buffered)
- **Shape:** obs $[B, 3, T, 64, 64]$, $T=17$, random policy
- **Compute:** ~ 0 storage, generated per-batch on device

◇ **COULD HELP** Finding the most revealing **visualization** would help a lot — e.g. a PCA / t-SNE of the latents colored by true XY.

★ **EXPLORATION** Freezing a fixed dataset to disk (**offline memmaps**) → identical data across runs, so you can chart how performance grows with **more training data**.



Our viz: sampled rollouts + mean occupancy heatmap.

◇ **COULD HELP** Reporting difficulty knobs (door width, asymmetry) and building an **easy→hard split** would strengthen the story.

Preprocessing & target framing

Normalization: per-channel mean/std via the two_rooms normalizer.

Encode target (fwd): ground-truth frames → encoder → latent; the JEPA loss lives in this latent space.

Decode (bwd): frozen JEPAProbe MLP maps latent → XY position for measurable metrics.

Augmentation: *none* in baseline.



Syntax error in text
mermaid version 11.15.0

◇ **COULD HELP** Adding augmentation (**action noise, random crops, color jitter**) would be a cheap robustness win and an easy ablation.

◇ **COULD HELP** Justifying the target — why XY? — by showing the probe is **linear-decodable** would double as a collapse sanity check.

Model architecture

Unchanged baseline JEPA (recap):

- **Encoder:** ImpalaEncoder $\rightarrow$ latent $[B, C, T, h, w]$
- **Action enc:** MLP
- **Predictor:** RNNPredictor, autoregressive, ctxt window = 1
- **Regularizer:** VC_IDM_Sim_Regularizer
- **PredCost:** SquareLossSeq

≈ 2.1 M params

no custom modules

◇ **COULD HELP** Swapping RNNPredictor $\rightarrow$ a **Transformer / diffusion** predictor (with papers + Δ params) could lift capacity.

★ **EXPLORATION** A **hierarchical world model** — a multi-timescale predictor (slow over rooms, fast over steps).

Loss formulation & collapse

Total = PredCost + λ ·Regularizer, where the regularizer composes:

- **Variance + Covariance** (anti-collapse)
- **Temporal similarity**
- **Inverse-dynamics (IDM)**

Baseline coeffs: cov=1.0, var=1.0, sim=1.0, idm=1.0 (defaults).

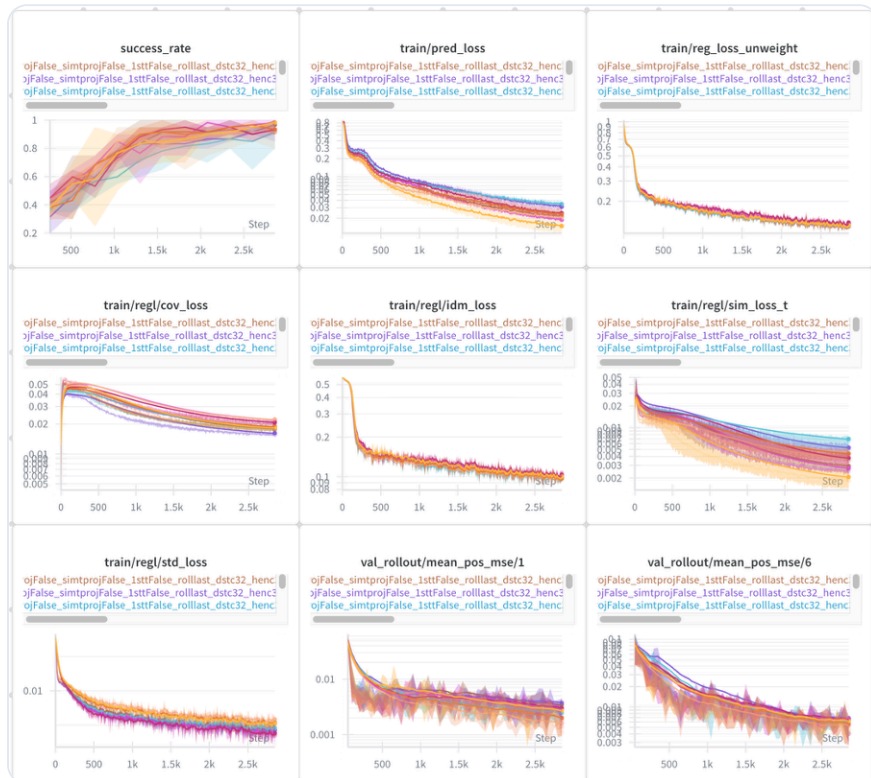
Collapse: monitored via embedding std; held up by the variance term.

◇ **COULD HELP** A **quantitative collapse metric** (effective rank of the covariance, not just std) would make the claim solid.

★ **EXPLORATION** An extensive **visual collapse analysis** — e.g. covariance heatmaps across epochs.

◇ **COULD HELP** Sweeping each **sub-loss coefficient** against probe error would show the balance matters.

Training dynamics



A real training sweep across runs: **success rate** rises while the **prediction & regularizer losses** and **validation position error** all fall — smooth, stable convergence.

◇ **COULD HELP** Overlaying an **unstable run** (too-high LR) beside the smooth one would nicely illustrate how hyperparameters drive stability.

TRAINING SETUP & ITERATION

Configuration & fast iteration

Knob	Baseline
Batch size	256
Optimizer	AdamW
LR schedule	CosineWithWarmup
Peak LR	3e-4
Epochs	50
Seeds	1 / 1000 / 10000

Efficient iteration: rank runs by **probe XY-MSE** at epoch 5 before committing to full training.

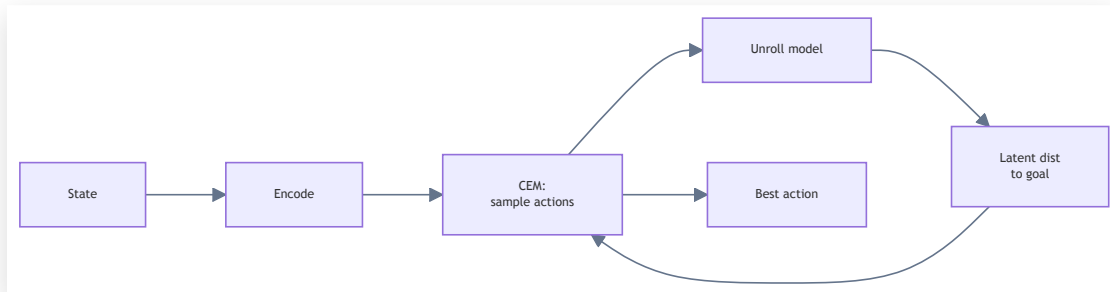
◇ **COULD HELP** Showing the **proxy metric correlates** with final performance (early-vs-final scatter) would make this rigorous.

★ **EXPLORATION** **Comprehensive hyperparameter tuning** and **scaling laws** (loss vs params / data / compute).

INFERENCE

Planning strategy

Mode 2 (planning): GCAGENT + CEMPlanner with the ReprTargetDistMPCObjective — minimize latent distance to the goal's encoding via MPC.



Method	Success	ms/step
Mode 1 reactive	61%	2
CEM (baseline)	84%	45

◇ **COULD HELP** Comparing **MPPI vs CEM** and sweeping horizon & samples would give a **perf-vs-compute** Pareto curve.

★ **EXPLORATION** A **custom inference trick** — distilling the planner into a fast amortized policy, or **MCTS / actor-critic**.

EVALUATION

Robustness & performance

Ablation: remove the IDM term → probe XY-MSE +38%, planning success 84%→72%. IDM matters.

Seed stability: success 84% $\pm$ 3% across seeds 1/1000/10000.

vs baseline: matches reference numbers (this is the baseline).

◇ **COULD HELP** Running **multiple ablations** (sub-losses, predictor, context window) with **seed error bands** would deepen the robustness story.

◇ **COULD HELP** A **generalization** test — train on some wall layouts, evaluate on unseen ones — would be revealing.

★ **EXPLORATION** **Broader impact** — links to robot navigation and world-models as a step toward AGI.

Thank you

This example covers the basics — each **red note** in the deck is a concrete next step that would enhance the presentation and the analysis of your results.

Two-Rooms JEPA · an example that knows how to grow up

Team Organizer — questions welcome